

Updated employment factors and occupational shares for the energy transition

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ABSTRACT

This study contributes to global energy workforce planning by presenting updated employment factors and occupational shares for the electricity workforce, covering generation, storage, and transmission construction. Global labour shortages are already recognised as a risk to energy sector transformation. To enable effective workforce development, it is important to know the scale, location, timing, and occupational composition of the required labour. However, data gaps may mean projections do not reflect current employment intensities and only provide gross numbers. A suite of indicators based on large-scale surveys in Australia and information from literature have been updated using cost declines as a proxy for labour productivity gains. This research presents employment factors for the number jobs created at each project stage, compares them to international estimates, and presents detailed occupational shares for five technologies. These indicators were used to make projections for the Australian 2024 Integrated System Plan for electricity, and sample results are discussed. These show that projected employment for large scale renewable generation under least cost scenarios for the energy transition are highly volatile. There is also significant overlap between occupations needed for different technologies and transmission. As countries try to rapidly expand renewable capacity and transmission infrastructure, integration of workforce projections into electricity scenarios using employment indicators allows for better planning. Workforce projections can inform policymakers of the need to reduce volatility, as well as helping training institutions, industry and governments to plan for the scale of workforce growth needed.

Abbreviations

ABS	Australian Bureau of Statistics
AEMO	Australian Energy Market Operator
EPC	Engineering, procurement and construction
FTE	Full-time equivalent
O&M	Operations & maintenance
PV	Photovoltaic
Units	
MW/MWh	Megawatt/Megawatt Hours
Jobs/MW	Full time equivalent employment created per MW of infrastructure (ongoing work)
Job-years/MW	Full time equivalent of one person working for a year per MW of infrastructure (this could be two people working for six months, or one person working for two years).

1. Introduction

To accelerate and plan for the energy transition, it is necessary to

predict the required energy sector workforce. This includes the construction of transmission lines, as a large expansion in transmission infrastructure will be needed to facilitate an electricity grid largely powered by renewable energy [1]. There is a lack of data on the number and location of energy sector jobs in most countries, and even less information on specific occupations [2]. There is almost no data on transmission construction employment. Data on the demand side workforce (energy efficiency, energy management, and electrification) is even scarcer [2]. International frameworks used by national statistics offices generally enable employment within fossil fuel and hydropower generation and fossil fuel extraction to be identified and analysed, while renewable generation is usually classified as ‘other electricity generation’. To further complicate matters, the nature of renewable projects means that a high proportion of jobs are classified under construction rather than power generation in census data. Transmission line construction is highly likely to appear in census returns for construction, with no means of separate identification. This results in a lack of robust empirical data for the world’s current energy sector, and hampers

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Table 1
Survey coverage by sector.

		Coverage		Additional information
Large-scale solar	Development and construction	1148	MW	45 % of industry in survey year
	Operations and maintenance	2063	MW	51 % of industry in survey year
Rooftop solar	Development and construction	253	MW	13 % of industry in survey year
	Operations and maintenance	658	MW	9 % of industry in survey year
Wind (onshore)	Development and construction	1468	MW	87 % of industry in survey year
	Operations and maintenance	6331	MW	74 % of industry in survey year
Hydro	Operations and maintenance	7522	MW	97 % of industry in survey year
Small-scale batteries	Development and construction	9.7	MW	21 % of residential batteries
	Operations and maintenance	35.7	MW	40 % of residential installations
Transmission lines	Double circuit	1907	km	6 responses
	Single circuit	547	km	2 responses
Transmission projects (non-line)	High and medium voltage	697	Million \$ (Australian)	7 responses

projection of the future workforce.

The University of Technology Sydney (UTS) undertook the first large-scale renewable energy industry survey for Australia in 2020 and carried out a similar process for transmission construction companies in 2021. The primary aim of both surveys was to derive energy sector employment indicators. The 2020 survey was funded by the Clean Energy Council and the 2021 survey by Infrastructure Australia. An additional literature review was conducted in 2022 to improve the employment indicators for utility-scale batteries, as the battery industry was at a very early a stage of development in 2020, with insufficient installations of large-scale batteries to derive robust indicators. This study describes the derivation of the employment factors and examines some of the results and their implications for workforce planning.

In 2015, UTS research presented a suite of global employment indicators, reflecting the current state of knowledge [3]. These employment factors have been used in the last five years [4–7] although indicators have changed significantly as renewable industries have matured. This research presents a comprehensive update, with the addition of several technologies (offshore wind, batteries, and transmission line construction). It is important that modellers are able to access consistent sets of indicators and update them to reflect the scenario time period.

The employment indicators published previously covered only the gross employment per megawatt (MW) or megawatt hours (MWh), which allowed comparisons of overall energy sector job growth or losses per technology. Although useful, this will not assist greatly in planning for the workforce, as information is needed on specific occupational changes.

This publication presents detailed occupational shares for seven technologies for the first time. Shares other than for coal and gas were derived in the 2020 and 2021 technology surveys, which collected detailed occupational shares at each stage of development as well as gross job creation, considerably extending the length and complexity of the interviews.

The derived indicators allow for analysis of the occupational composition of the workforce and the overall quantum of employment by applying the indicators to electricity sector scenarios. The resulting projections can be used to inform training requirements, thus facilitating a smooth transition.

Technologies for which this research offers updated or initial employment factors include:

- Utility-scale and rooftop solar – gross employment updated, occupational shares presented in detail
- Onshore wind – gross employment updated, occupational shares presented in detail
- Hydro and pumped hydro generation – gross employment updated, occupational shares presented in detail
- Transmission line construction – occupational shares presented in detail
- Battery storage – gross employment updated, occupational shares not available
- Offshore wind – gross employment updated, occupational shares not available.

The employment factors presented included direct jobs in development and construction or installation, in operation and maintenance, and in manufacturing. The employment factors do not include induced jobs, such as the expenditure of solar or wind farm construction workers in regional towns for accommodation or food, or the general wage expenditure associated with employment increases.

2. Material and methods

Industry surveys were used to collect data on the overall employment, occupational composition, and skill shortages and recruitment issues within each sector. Survey data, supplemented by information from previous work and other literature, was then used to calculate employment factors in job-years or jobs per MW (for generation and batteries) and per kilometre or per million dollars (Australian) for transmission construction projects. These indicators were combined with electricity sector scenarios produced by the Australian Energy Market Operator (AEMO) to project employment in the generation and transmission construction sectors up until 2050.

The surveys collected data from companies involved in solar (large- and small-scale), wind, hydro, battery storage (large- and small-scale), and transmission construction. The surveys focussed on individual projects for large-scale construction, and total installations over a single year for small-scale solar and batteries. Information was also collected on cumulative installations for operational employment.

2.1. Survey coverage

The 2020 renewable technology surveys covered a large proportion of the Australian solar, wind, battery storage and hydro sectors, with a total of 268 valid responses. The 2021 transmission survey covered the main construction contractors for transmission infrastructure, including approximately 2500 km of line construction and just under \$700 million (Australian) worth of non-line construction projects. Table 1 summarises the survey coverage by industry sector and project phase.

Respondents covered 87 % of 2020 new wind capacity and 46 % of new solar capacity (74 % and 51 % of cumulative installed capacity respectively). The coverage for distributed solar was much lower (13 % of installations for the survey year), reflecting the smaller company size for the distributed solar sector and the greater difficulty in reaching relevant companies. The surveys covered nine major transmission projects, with 2454 km of line and \$697 million (Australian) capital expenditure on non-line projects. For more details on the survey methodology and outcomes see Briggs et al., 2022 [8].

The disaggregated nature of the industry makes collecting

comprehensive data challenging. Construction involves many layers of subcontracting, and few wind farm development companies have oversight of employment which occurs in contractor or subcontractor companies. The survey asked respondents to indicate the proportion of subcontract employment they were able to identify. Where possible, the project team followed up with appropriate subcontractors to fill in information, particularly where the gap covered major aspects of employment. While new transmission lines will certainly require operations and maintenance, it was anticipated that this would be absorbed within the normal business of transmission companies.

2.2. Calculation of employment factors – overview

Employment factors were derived using a staged process for all technologies.

- Functional units were derived for each generation technology and for transmission line construction to guide total employment intensity data collection. Most functional units derived aligned with previous literature sources and standard industry practice. For example, the functional unit for wind tower construction was determined to be total MW of installed capacity. However, transmission lines required a different characterisation, resulting in a distance based functional unit (kilometres of line).
- A list of key occupations was developed and sense-tested with experts in the field before developing the industry surveys.
- A survey was designed to collect employment intensity data for each project phase and technology.
- The average of all projects in each technology category was calculated to derive a final employment factor in FTE jobs or job-years/functional unit.

The resulting employment factors (jobs or job-years/functional unit) are multiplied by the gross number of installed functional units for a given year to produce time series modelled projections per technology. For example, an employment factor of X job-years/MW of installed capacity for construction is multiplied against the total MW of installed capacity due to be constructed for that year. When the construction takes more than a single year, the employment factor is divided by the construction time and represents the average employment over the construction period. When looking at specific projects, rather than regionally or nationally, employment peaks within the construction period will be important.

2.3. Calculation of gross employment factors for renewable energy & batteries

The same core methodology is used to derive employment factors for generation technologies and for batteries, with only minor variations. The employment factor was derived using a weighted average of survey responses in each of the project phases by summing the employment and dividing by the total capacity managed or installed in MW. The employment factor therefore gives a greater weighting to data from respondents with multiple or larger projects.

For large-scale solar and wind, the construction and operations and maintenance phases were subdivided for the calculation of employment factors to reflect industry subdivision into EPC (engineering, procurement and construction) management companies that oversee and organise work, and EPC companies that undertake the actual work. Separate employment factors were calculated for management and actual implementation, to allow the maximum data coverage of the survey and avoid distortions as the contracting entities (the EPC management companies) have very small employment footprints relative to the implementing companies (the EPC firms).

The battery surveys were undertaken in 2020, with 39 responses in total. While the estimated coverage of the residential market was good,

at approximately 20 %, the surveys did not have respondents who had experience of installing grid connected batteries. At the time there were only seven utility batteries operational in Australia with a combined capacity of just under 300 MW [9], compared to approximately 90,000 distributed batteries with capacity around 500 MW [10] (assuming an average capacity of 11 kWh and 2.2 kW). Most of the battery survey respondents were companies that also installed distributed solar PV. This study used the commercial and industrial battery results for utility battery employment in the 2020 workforce projections in the absence of better data [11], although this was likely to overstate the employment creation. The combined employment factor for distributed batteries was heavily weighted towards residential batteries, reflecting their relative capacities. Residential batteries accounted for 85 % of installations and 96 % of operations and maintenance activity among respondents.

The need for better employment factors for utility batteries was partially addressed in 2022, when undertaking energy workforce projections to accompany AEMO's Integrated System Plan. The survey process was unable to be repeated, so a literature search was undertaken to identify both international estimates of battery employment and collate information from Australian projects. This identified thirteen Australian utility batteries projects, two operating, five at scoping stage, and six approved, with useable employment data. These had a combined capacity of 3.3 GW, an average capacity of 235 MW, and weighted average storage time of 3.2 h. All projects gave information on installation employment, and 0.9 GW included operations and maintenance employment. The weighted average installation employment was 0.61 job-years/MW capacity, with 0.04 jobs/MW in operations and maintenance [12].

2.4. Calculation of gross employment factors for transmission construction

The Australian Bureau of Statistics (ABS) publishes occupational data for the electricity transmission sector every five years. However, this data represents workforce requirements for the maintenance and scheduled upgrades of transmission infrastructure rather than the workforce required to build new transmission infrastructure. Future construction is largely greenfield, which involves securing and clearing new access corridors, building large interconnectors between states, and laying down some of the largest substations and non-line assets that have ever been built in Australia.

To fill the data gap, two options were available, either to use employment factors and occupational breakdowns from the literature or to develop employment factors using surveys. It is important that employment factors from the literature are geographically suitable, and occupational breakdowns are required to understand the needs of the future workforce.

Several sources estimated the employment intensities for proposed transmission line build, however the majority relied on macro-level modelling methods such as input-output analysis which calculates employment in relation to project expenditure. This is appropriate in the absence of better data, but less accurate than surveys due to the considerable uncertainties that price volatility introduces. Most of the studies lacked data on employment intensity for certain asset types and most sources lacked detailed occupational shares. Surveys were therefore undertaken in 2021 to derive employment intensities and the corresponding occupational shares.

Desktop research informed the design of the transmission employment survey. Firstly, data from literature was used to inform the functional units for employment intensities for both line and non-line assets. A functional unit is required that is sufficiently detailed to reflect variations in employment intensity across asset types. The classification of transmission infrastructure was particularly challenging due to the large number of asset types and sub-types. It was unclear at the stage of survey design whether variations in line voltages, number of circuits, height of transmission towers, size of substations, or geographical location would

Table 2

Decline factors used to adjust suite of indicators presented.

	2022	2023	2024
Utility-scale PV	100 %	84 %	78 %
	100 %	92 %	84 %
Distributed batteries			
	100 %	90 %	79 %
Utility-scale batteries			
	100 %	99 %	99 %
Wind (onshore)			
	100 %	97 %	95 %
Wind (offshore)			
	100 %	88 %	80 %
Rooftop PV			

result in differences in employment intensities. Initially a detailed classification framework was developed including various line attributes, for example tower height and type, tower spacing, whether single or double circuit. Terrain attributes were also collected (forested, mountainous, flat).

Detailed survey data was collected according to line and non-line asset type and transmission attributes via nine surveys with Transmission Network Service Providers and their EPC contractors across the National Electricity Market member states. Two functional units were commonly used by industry when referring to employment intensities, length of the transmission line in kilometres, and the total value of the project in millions of dollars (Australian) for non-line transmission projects. Detailed employment data was collected for 23 transmission projects using these two functional units [25].

Variations in employment intensities and occupational composition was measurable between single and double circuit lines, and high and low voltage non-line assets. These parameters were used to differentiate employment intensities. Terrain type was reported by survey participants to play a notable role in terms of extending project timeframes and requiring specialised skills. However, survey data did not show a consistently different employment intensity according to terrain type and so these impacts were averaged over all asset types while deriving employment factors [25].

Data on Full-Time Equivalent (FTE) jobs, occupational share per project phase, and project duration was collected from survey

participants and combined to calculate final employment intensities in total FTE/km and FTE/\$million. Survey data was collected by project phase (development and construction), and then aggregated into a single development and construction indicator. Operations and maintenance employment was not included. More detail on the survey methodology can be found in Briggs et al., 2021 [13].

2.5. Occupational shares

Survey participants provided information on total employment intensity for each technology and then proceeded to break down total employment by occupation. Occupational shares are presented as a percentage of total employment for each technology and project phase. It was not feasible to obtain breakdowns for all technologies, with exclusions as follows:

- Grid scale and residential battery installation and operations and maintenance; the number of responses was insufficient.
- Transmission operations and maintenance.
- Coal and gas occupational shares were calculated from ABS data rather than from surveys, and are only available for operations and maintenance and fuel supply.

Occupational classification survey respondents were asked to categorise employment for each development phase and type of project using both the Australian and New Zealand Standard Classification of Occupations (ANZSCO) 1-digit occupational categories and a tailored list of occupations. The occupation list was determined in industry consultations and corresponded to 2-digit, 3-digit or 4-digit ANZSCO categories; a blended list was required in order to cover the main occupations of relevance to the specific projects. Respondents were also asked to allocate the percentage of local (defined as within approximately 100 km of the project site) employment corresponding to each occupation.

2.6. Decline factors

Decline factors are applied to the employment factors to account for increases in labour productivity over time and avoid over-estimating future employment. There are two major sources of increased labour

Table 3

Employment factors for electricity generation technologies, storage, and transmission construction.

	Decline applied	Construction/installation	Manufacturing		O&M	Fuel
			Total	Australian		
			Job-years/MW		Jobs/MW	Jobs/GWh
Black coal ⁽¹⁾	100 %	11.08	5.41	1.6	0.22	0.04
	100 %	11.08	5.41	1.6	0.22	0.01
Brown coal ⁽¹⁾						
	100 %	1.27	0.92	0.28	0.14	0.07
Gas ⁽¹⁾						
	100 %	7.36	2.3	1.04	0.14	
Hydro ^(1,2)						
Onshore Wind ^(1,2)	99 %	2.65	1.54	0.37	0.21	
Offshore Wind ⁽³⁾	n/a	1.5	13.68	0.90	0.28	
Utility solar ^(1,2)	78 %	1.61	3.08	0.07	0.09	
Rooftop solar ^(1,2)	80 %	4.19	2.86	0.12	0.13	
Utility batteries ⁽¹⁾	88 %	0.53	0.31	0.08	0.03	
Distributed batteries ^(1,2)	84 %	4.44	0.31	0.08	0.23	
Pumped hydro ^(1,2)	100 %	7.18	6.4	0.70	0.08	
Transmission lines ^(1,2)		Construction/installation (Job-years/km)				
Single circuit	100 %	0.7				
Double circuit	100 %	3.7				
		Construction/installation (Job-years/\$m)				
Transmission (other) ^(1,2)		1.9				

Note 1: Employment factors from Rutovitz et al., 2023 [12] with construction factors updated using the listed decline factors.

Note 2: Employment factors derived via surveys, noting that international manufacturing factor is from literature in all cases.

Note 3: Offshore wind factors from Rutovitz et al., 2024 [33].

Table 4

Comparison of current employment factors to previous work and other sources.

	Project phase	Unit	This research	UTS 2022 [12]	UTS 2020 [21]	UTS 2015 [3]	Additional sources/notes
Batteries (distributed)	Development and construction	Job-years/MW	4.44	5.3	5.6	–	Literature estimates for installation and operations and maintenance not available
	Operations & maintenance	Jobs/MW	0.23	0.3	0.3	–	
Batteries (distributed and utility)	Manufacturing (total)	Job-years/MWh	0.11 [22]	–	–	–	0.11 Steen et al., 2017 [23]
							0.35 Nevada Governor's Office 2018 [24]
							0.18 Faraday Institute, 2020 [25]
							0.14 Thielmann et al., 2021 [26]
							0.2 Manager Magazin, 2021 [27]
							0.42 BVES, 2022 [28]
Batteries (utility)	Manufacturing (total)	Job-years/MW	0.31	0.6	6.6	–	0.12 Usai et al., 2022 [29]
							0.16 Auto Motor Sport, 2023 [30]
							112 job-years/GWh are converted to job-years/MW using the ratio 2.7MW/MWh [22]
Solar (rooftop)	Development and construction	Job-years/MW	4.2	5.2	5.8	13	4.7 Australian Bureau of Statistics (2019, pers. comm), assuming 6 kW per installation
							8.1 Ernst & Young 2017 [31]
Solar (large-scale)	Manufacturing (total)	Job-years/MW	2.9	3.6	4.4	6.7	Calculated from National Solar Jobs Census 2022 and Solar Energy Industries data [32]
							4.2 IRENA 2017 [33]
Hydro	Manufacturing (total)	Job-years/MW	0.12	0.15	0.15	0.7	4.7 IRENA 2017 [33]
Pumped hydro	Manufacturing (total)	Job-years/MW	0.13	0.16	0.16	0.7	3.45 IRENA 2017 [33]
Wind (onshore)	Manufacturing (total)	Job-years/MW	0.09	0.09	0.11	–	4.2 IRENA 2017 [33]
Wind (offshore)	Manufacturing (total)	Job-years/MW	0.09	0.09	0.11	–	1.13 IRENA 2017 [33]
							0.12 Solar Energy Industries Association 2023
							8.7 Navigant/NREL 2017 [34]
Wind (onshore)	Manufacturing (total)	Job-years/MW	0.09	0.09	0.11	–	2.3 NREL 2016 [35]
Wind (offshore)	Manufacturing (total)	Job-years/MW	0.14	0.1	0.14	0.2	0.25 Navigant/NREL 2017 [34]
							0.23 NREL 2022 [36]
							17.2 NREL 2016 [35]
							4.3 Biggar Economics [37]
							6.4 NREL 2016 [35]
Wind (onshore)	Manufacturing (total)	Job-years/MW	0.08	0.1	0.08	–	0.23 NREL 2022 [36]
							0.06 Biggar Economics [37]
							0.09 NREL 2022 [36]
							1.2 ABS (2019, pers. Comm)
							3.1 IRENA 2017 [38]
							1.6 IRENA 2018 [39]
Wind (offshore)	Manufacturing (total)	Job-years/MW	0.21 ⁽³⁾	0.2	0.22	0.3	0.23 Calculated from GWEC 2022 [40]
							0.09 ABS (2019, pers. Comm)
							1.65 IRENA 2017 [39]
							2.2 IRENA, 2018 [39]
							1.4 Sylvest, 2020 [41]
							0.4 Knol and Coolen, 2020 [42]
Wind (offshore)	Manufacturing (total)	Job-years/MW	13.68	5.3	–	15.6	10.5 IRENA, 2018 [39]
							5.3 Sylvest, 2020 [41]
							3.3 Knol and Coolen, 2020 [42]
							0.17 IRENA, 2018 [39]
							0.08 Sylvest, 2020 [41]
							0.07 Knol and Coolen, 2020 [42]

productivity which could decrease the employment factor in practice. Firstly, there could be capital deepening. For example, a major driver of cost decline in wind technology is turbine size. The average turbine size in Australian wind farms is 5–6 MW per turbine [14], compared to 1.3

MW for wind farms commissioned between 2001 and 2005 [26]. Whilst larger turbines will require some additional installation effort, this is not proportional to the increased capacity [15], so the employment factor declines as a result. Secondly, there could be increased labour

Table 5

Occupational shares, development and construction phase.

		Onshore wind	Utility-scale PV	Rooftop PV	Hydro	Pumped hydro	Transmission		
							Line: double	Line: single	Other
Gross employment intensity (job-years/MW for generation, job-years/km lines, job-years/\$m other)		2.7	1.6	4.2	7.4	7.2	3.7	0.7	1.9
Occupational shares development and construction phase (%)									
MANAGERS	Executives & senior managers	5.3 %	2.0 %	5.4 %	1.2 %	1.2 %	1.3 %	1.8 %	–
	Business development managers	0.8 %	0.1 %	3.6 %	2.1 %	2.1 %	–	–	–
	Business managers	5.0 %	3.4 %	2.8 %	0.7 %	0.7 %	0.7 %	–	–
	Construction managers	9.6 %	7.2 %	3.7 %	1.8 %	1.8 %	2.6 %	7.2 %	8.2 %
	Engineering managers	–	–	–	–	–	0.9 %	0.4 %	2.1 %
	Operations & production managers	–	–	–	0.1 %	0.1 %	–	–	–
	Logistics & procurement managers	4.6 %	1.6 %	2.3 %	0.4 %	0.4 %	0.7 %	–	–
PROFESSIONALS	Finance, business, legal & policy professionals	5.8 %	1.3 %	11.4 %	2.2 %	2.2 %	–	–	–
	Professionals (other)	4.0 %	0.6 %	5.9 %	1.6 %	1.6 %	2.2 %	0.3 %	0.5 %
	Civil engineers	2.7 %	2.1 %	–	5.9 %	5.9 %	1.0 %	0.9 %	2.4 %
	Geotechnical engineers	–	–	–	–	–	0.2 %	–	0.7 %
	Quantity surveyors	–	–	–	–	–	0.1 %	0.1 %	0.8 %
	Structural engineers	–	–	–	–	–	1.8 %	0.6 %	2.2 %
	Electrical engineers	8.1 %	3.0 %	1.5 %	2.7 %	2.7 %	2.1 %	12.6 %	9.0 %
	Mechanical engineers	1.0 %	1.6 %	–	2.7 %	2.7 %	–	–	–
	Other engineers	–	–	–	–	–	0.4 %	–	0.2 %
	Community engagement	2.1 %	–	–	0.9 %	0.9 %	1.3 %	–	0.5 %
	Environment, health & safety	6.4 %	3.5 %	1.1 %	3.1 %	3.1 %	1.0 %	1.0 %	2.1 %
	IT professionals	–	–	1.7 %	–	–	0.3 %	–	–
	Telecommunications engineer	1.5 %	0.3 %	–	0.4 %	0.4 %	0.3 %	–	0.8 %
TRADES & TECHNICIANS	Trades & technicians (other)	3.6 %	7.7 %	5.4 %	3.0 %	3.0 %	7.7 %	2.0 %	5.0 %
	Civil engineering technicians	–	–	–	2.7 %	2.7 %	–	–	–
	Mechanical trades & technicians	4.5 %	6.9 %	0.2 %	–	–	–	–	2.6 %
	Metal trades	0.5 %	–	–	3.2 %	3.2 %	–	–	–
	Construction trades	0.3 %	0.8 %	–	–	–	–	3.3 %	–
	Plumbers	0.2 %	0.4 %	–	–	–	–	–	–
	Electricians	7.0 %	15.9 %	17.9 %	2.5 %	2.5 %	–	–	24.0 %
	Lineworkers	–	–	–	–	–	27.5 %	36.7 %	1.6 %
	Cable jointers	–	–	–	–	–	–	0.5 %	0.4 %
ADMIN	Telecommunications technician	0.3 %	–	–	0.4 %	0.4 %	–	–	0.2 %
	Admin staff	4.8 %	9.3 %	8.0 %	2.9 %	2.9 %	4.8 %	5.1 %	3.9 %
MACHINE OPERATORS & DRIVERS	Crane operators	2.9 %	–	–	1.0 %	1.0 %	1.8 %	1.1 %	0.8 %
	Machine operators & drivers (other)	–	–	0.5 %	–	–	3.7 %	3.5 %	0.3 %
	Drillers	1.0 %	–	–	–	–	0.3 %	0.9 %	2.1 %
	Earthmoving plant operators	4.7 %	1.3 %	–	1.8 %	1.8 %	5.1 %	3.8 %	7.6 %
	Truck drivers	1.6 %	6.3 %	3.7 %	9.3 %	9.3 %	7.7 %	2.0 %	1.6 %
	Storepersons & packers	–	0.9 %	3.4 %	–	–	–	–	–
LABOURERS	Construction labourers	1.6 %	22.6 %	1–	21.1 %	21.1 %	4.2 %	3.1 %	3.7 %
	Concreters	3.2 %	–	–	17.7 %	17.7 %	0.8 %	2.3 %	4.9 %
	Riggers & dogmen	2.6 %	1.4 %	–	2.7 %	2.7 %	16.1 %	5.9 %	1.9 %
	Steel fixers	1.8 %	–	–	3.5 %	3.5 %	3.6 %	3.6 %	3.3 %
	General labourers	0.4 %	–	0.5 %	–	–	–	0.2 %	–
	Electrical trade assistants	2.1 %	–	10.9 %	2.7 %	2.7 %	–	1.1 %	5.8 %
	Telecommunications trade assistant	–	–	–	–	–	–	–	0.1 %

productivity from more efficient processes or technological change. For example, solar farm construction times have reduced from learning-by-doing. Automation is expected to reduce labour in areas such as panel cleaning. Learning rates are a well-established feature of studies on renewable energy cost declines [16].

Broadly, there are two major options to be used for a decline factor:

- **Historical trends in labour productivity:** changes in labour productivity over time could be applied to forward years. However, historical data on labour productivity for renewable energy does not exist, due in part to the industry's recent emergence and the absence of a good match with industry classifications in the Australian and New Zealand Standard Industry Classification (or the International Standard Classification). Renewable energy construction, for example, is embedded within heavy civil engineering construction which would require using a factor which combines trends from a range of construction activities.

- **Projected declines in capital cost (or 'learning rates'):** as renewable energy technologies have exhibited large reductions in capital cost (\$/MW), there is a significant body of literature on learning rates, and it is commonplace to apply decline rates to capital cost projections. In the Australian case, the Commonwealth Scientific Industrial Research Organisation (CSIRO) develops estimates annually based on a combination of learning-by-doing rates, international trends and current trends which are applied by the AEMO in its modelling (the GenCost series).

Whilst changes in the capital cost of renewable energy technologies are not an exact proxy as they will include other variables unrelated to labour productivity, our view is that projections based on trends and practices within the renewable energy sector are likely to be more accurate than using a historical labour productivity in an associated sector such as heavy civil engineering construction.

Table 6

Occupational shares, operation and maintenance phase.

	Onshore wind	Utility-scale PV	Rooftop PV	Hydro	Pumped hydro	Coal	Gas
Gross employment intensity (jobs/MW)	0.21	0.09	0.13	0.14	0.08	0.22	0.14
Occupational shares O&M phase (%)							
MANAGERS							
Executives & senior managers	1.6 %	12.3 %	5.4 %	1.7 %	1.7 %	1.1 %	1.1 %
Business development managers	–	–	3.6 %	2.4 %	2.4 %	0.6 %	0.6 %
Business managers	–	9.2 %	2.8 %	10.3 %	10.3 %	2.0 %	2.0 %
Construction managers	–	–	3.7 %	1.3 %	1.3 %	0.6 %	0.6 %
Engineering managers	–	–	–	2.6 %	2.6 %	1.1 %	1.1 %
Operations & production managers	11.4 %	14.5 %	–	1.4 %	1.4 %	0.8 %	0.8 %
Logistics & procurement managers	–	–	2.3 %	–	–	0.5 %	0.5 %
Managers (other)	–	–	–	–	–	5.6 %	5.6 %
PROFESSIONALS							
Finance, business, legal & policy professionals	9.5 %	2.8 %	11.4 %	9.9 %	9.9 %	3.0 %	3.0 %
Professionals (other)	–	–	5.9 %	2.1 %	2.1 %	3.0 %	3.0 %
Civil engineers	–	–	–	2.6 %	2.6 %	0.6 %	0.6 %
Electrical engineers	–	–	1.5 %	7.5 %	7.5 %	3.5 %	3.5 %
Mechanical engineers	–	–	–	2.7 %	2.7 %	2.5 %	2.5 %
Other engineers	–	–	–	1.4 %	1.4 %	0.9 %	0.9 %
Community engagement	1.7 %	–	–	–	–	–	–
Environment, health & safety	4.3 %	1.9 %	1.1 %	1.4 %	1.4 %	1.8 %	1.8 %
IT professionals	–	–	1.7 %	–	–	1.5 %	1.5 %
Telecommunications engineer	–	–	–	7.5 %	7.5 %	0.2 %	0.2 %
TRADES & TECHNICIANS							
Trades & technicians (other)	–	–	5.4 %	–	–	8.6 %	8.6 %
Civil engineering technicians	–	–	–	2.7 %	2.7 %	–	–
Mechanical trades & technicians	22.7 %	12.3 %	0.2 %	5.3 %	5.3 %	0.1 %	0.1 %
Metal trades	–	–	–	5.5 %	5.5 %	12.6 %	12.6 %
Construction trades	–	–	–	–	–	0.2 %	0.2 %
Plumbers	–	–	–	–	–	0.4 %	0.4 %
Electricians	4-	31.3 %	17.9 %	6.4 %	6.4 %	8.7 %	8.7 %
Lineworkers	–	–	–	–	–	1.6 %	1.6 %
Telecommunications technician	–	1.7 %	–	1.7 %	1.7 %	1.1 %	1.1 %
Control room & power plant operators	2.1 %	4.5 %	–	7.0 %	7.0 %	13.5 %	13.5 %
ADMIN							
Admin staff	6.6 %	8.9 %	8.0 %	14.9 %	14.9 %	11.6 %	11.6 %
MACHINE OPERATORS & DRIVERS							
Crane operators	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.3 %	0.3 %
Machine operators & drivers (other)	0.0 %	0.0 %	0.5 %	0.8 %	0.8 %	1.9 %	1.9 %
Drillers	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	1.8 %	1.8 %
Earthmoving plant operators	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.7 %	0.7 %
Truck drivers	0.0 %	0.0 %	3.7 %	0.0 %	0.0 %	0.2 %	0.2 %
Storepersons & packers	0.0 %	0.0 %	3.4 %	0.0 %	0.0 %	0.9 %	0.9 %
LABOURERS							
Construction labourers	0.0 %	0.0 %	10.0 %	0.5 %	0.5 %	0.5 %	0.5 %
Steel fixers	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	2.4 %	2.4 %
General labourers	0.0 %	0.6 %	0.5 %	0.4 %	0.4 %	3.3 %	3.3 %
Electrical trade assistants	0.0 %	0.0 %	10.9 %	0.0 %	0.0 %	0.0 %	0.0 %

$$\text{Decline factor for year } X = \left(\frac{\text{technology cost (year } x)}{\text{technology cost (base year)}} \right)$$

The decline factor is used to calculate the relevant employment factor for each year:

$$\text{Employment factor (year } x) = \text{Employment factor (base year)} \times \text{decline factor (year } x)$$

It is particularly important to apply decline factors when using employment factors alongside energy projections as it is highly unlikely that the labour intensity of each technology will remain static while the technology cost reduces. In that case, it is advised to use the cost declines which are intrinsic to that energy projection.

The decline factor used to adjust the employment indicators to 2024 is based on the decline in capital cost for renewable energy and battery technologies used by AEMO for the 2022 Integrated System Plan, combined with the GenCost 2020-21 report [17,18].

The declines used are shown in Table 2, and vary from a maximum of a 20 % decline for rooftop solar to just 1 % decline for onshore wind.

Where technologies are not listed (for example, transmission construction or gas generation) no cost decline is used as the source data does not show a cost decline.

The impact of decline factors is significant even in the two years from 2022 to 2024. Although the decline factor itself usually reduces year by

year, the effect is cumulative and applying decline factors can see employment indicators reduce by sixty to seventy percent over a projection timescale of twenty years.

3. Results

3.1. Employment factors to calculate scale of workforce

Employment factors are presented in job-years/MW for construction and manufacturing phases, operations and maintenance employment factors are presented as jobs/MW and for fuel, jobs/GWh (Table 3).

Table 7
Occupational shares, fuel.

		Fuel	
		Coal	Gas
Gross employment intensity (jobs/GWh)		0.04 (black coal) 0.01 (brown coal)	0.07
Occupational shares for fuel (%)			
MANAGERS	Executives & senior managers	0.6 %	0.6 %
	Business development managers	0.3 %	0.3 %
	Business managers	1.4 %	1.4 %
	Construction managers	0.2 %	0.2 %
	Engineering managers	0.5 %	0.5 %
	Operations & production managers	2.6 %	2.6 %
	Logistics & procurement managers	0.3 %	0.3 %
	Managers (other)	0.8 %	0.8 %
PROFESSIONALS	Finance, business, legal & policy professionals	3.0 %	1.8 %
	Professionals (other)	1.1 %	1.7 %
	Civil engineers	0.2 %	0.2 %
	Electrical engineers	0.8 %	0.8 %
	Mechanical engineers	0.8 %	0.8 %
	Other engineers	2.1 %	2.1 %
	Environment, health & safety	1.0 %	1.0 %
	IT professionals	0.6 %	0.6 %
	Telecommunications engineer	0.1 %	0.1 %
TRADES & TECHNICIANS	Trades & technicians (other)	7.8 %	7.8 %
	Mechanical trades & technicians	0.1 %	0.1 %
	Metal trades	15.8 %	15.8 %
	Construction trades	0.2 %	0.2 %
	Plumbers	0.1 %	0.1 %
	Electricians	5.3 %	5.3 %
	Telecommunications technician	0.2 %	0.2 %
	Control room & power plant operators	0.1 %	0.1 %
ADMIN	Admin staff	5.2 %	5.2 %
MACHINE OPERATORS & DRIVERS	Crane operators	4.1 %	4.1 %
	Machine operators & drivers (other)	33.5 %	33.5 %
	Drillers	2.1 %	2.1 %
	Truck drivers	6.0 %	6.0 %
	Storepersons & packers	0.9 %	0.9 %
LABOURERS	Construction labourers	0.6 %	0.6 %
	Steel fixers	0.3 %	0.3 %
	General labourers	1.0 %	1.0 %

These are the indicators used to calculate the scale of the workforce including all occupations.

Construction employment factors other than for coal, gas, and offshore wind were derived from detailed survey data from 2020 to 2021, adjusted from the 2022 values using the decline factors shown (for details on how the decline factor was calculated see section 3.5). Declines are also applied to manufacturing.

Local manufacturing percentages were determined during the same surveys, except for offshore wind, which assumes 17.5 % occurs onshore, the average of the higher and lower figures in Briggs et al., 2021 [19]. Batteries manufacturing is the average value from eight studies that occurred between 2017 and 2022, with a range of 0.11 job-years/MWh to 0.42 job-years/MWh. This was converted to jobs-years/MW assuming an average value of 2.7 MW/MWh found in the fourteen projects used to derive the construction employment indicator.

There is a steady downward progression in the employment factors for newer technologies; for example, development and construction of rooftop solar declines from a calculated indicator of 13 job-years/MW in

2015 [18], to a figure of 5.8 job-years/MW calculated in the 2020 industry survey. Fossil fuels and hydro employment factors, by contrast, are not expected to decline.

The employment indicators for offshore wind shown in Table 3 are those derived in 2024 in Rutovitz et al. [32] using the National Renewable Energy Laboratory publicly available model for offshore wind [20] and making assumptions for local content. These values are not incrementally changed from values used in 2022, which were considerably lower, as shown in Table 4.

3.2. Comparison with previous work and other employment indicators

The updated employment factors listed in Table 4 are compared to previous work [3,12,21] and where possible, some alternative indicators from other studies. There is considerable variation in the indicators that are used. For example, the employment indicators for battery manufacturing ranges from 0.11 to 0.35 in the literature cited. This reflects the variation in factories, the methods for calculating the employment, and the situations considered. Employment per unit may be much higher at the start of a factory operation, simply because the factory is not producing at full capacity. Choosing which indicator to use requires judgement on how each indicator has been derived, and how applicable it is to the situation of concern. Another crucial consideration is how up to date it is, particularly when industries are relatively immature and employment creation is changing rapidly.

Within our own work, utility batteries and solar rooftop have seen stepwise reductions in employment indicators. These usually reflect the point when the indicator switched from literature derived to survey derived data based on industry respondents.

The variation in offshore wind employment indicators is significant and reflects not only the relative immaturity of the industry but variation in the activities included. The attribution of the employment associated with specialised vessel operation and maintenance, and the boundary of the direct supply chain results in significant variation in the manufacturing employment and in O&M employment. One interesting observation is that O&M employment includes a significant supply chain element.

3.3. Occupational employment shares

Occupational breakdowns for individual project phases are presented in Table 5 (development and construction), Table 6 (operations and maintenance), and Table 7 (fuel supply). These indicators are provided as percentages to be multiplied by the total employment by technology and project phase, or alternatively applied to the employment factors to gain occupational indicators.

For example, the total employment intensity per MW of installed capacity for rooftop PV over the construction phase is 5.8 job-years (Table 3) and the occupational share of 'executives and senior managers' for the construction phase of rooftop PV systems is 5.4 % (Table 5), which equates to 0.3131 job-years per MW.

3.4. Workforce projections - sample results

The employment factors and occupational shares presented were used to provide workforce projections alongside the Integrated System Plan for Australia's National Electricity Market [43,44]. The plan produces electricity scenarios every two years that meet projected electricity demand and government targets for decarbonisation. The modelling tends to defer capital expenditure as late as possible as it takes a least cost approach and does not currently impose limits on building each year. Legislated targets for 2030 lead to a construction boom in the latter part of the 2020s as so much infrastructure is required to decarbonise the system. This is repeated to a lesser extent towards 2050.

Fig. 1 shows the workforce projections for the central scenario of the plan, from 2024 to 2050. The figure shows the employment profile for

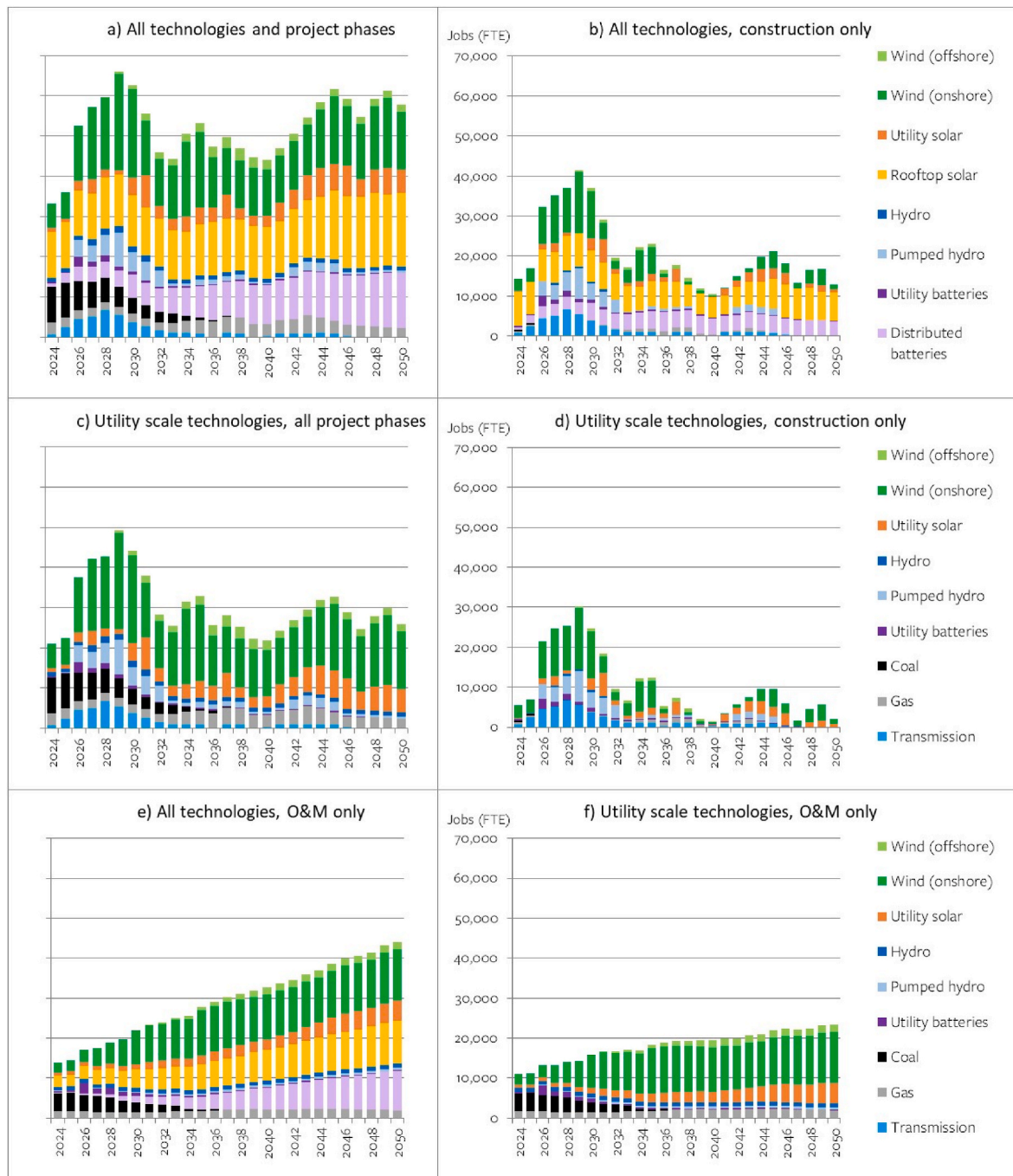


Fig. 1. Employment by technology for the Australian Integrated System Plan, Step Change scenario.

all project phases (a, c and e) and for construction only (b, d and f). The volatility is almost entirely driven by construction, as may be seen by each pair. Operations and maintenance (e and f) have a steady growth profile over the entire period. The workforce for utility-scale projects and rooftop solar and distributed batteries are unlikely to overlap to any great extent. The two distributed technologies have a profile of steady growth over the period and reduce the apparent volatility. The distributed workforce is removed from [graphs 1c and 1d](#), demonstrating that actual volatility is greater than shown, with a reduction in the total construction workforce of 25,000 between the peak year of 2029 and 2033 (graph d), compared to 17,000 when all phases and distributed technologies are included (graph a).

The peak year breakdown by occupation is shown in [Fig. 2](#) for the ten most in-demand occupations. These ten account for 55 % of the

calculated occupational employment, with the remaining 45 % spread over another thirty-five occupations. The lack of occupational shares for batteries and most manufacturing means that occupational shares are only available for 87 % of employment.

Electricians dominate by a significant amount, accounting for 13 % of calculated occupational employment in the peak year. [Table 8](#) gives the three most in-demand occupations for each technology in that peak year. Electricians are the most in demand for all the renewable generation technologies and come third for transmission construction. It is likely that electricians will also be in the top three for batteries, although that occupational breakdown is not available. The overlap of in-demand occupations between different technologies may make it harder to fill vacancies but may also make a career in renewable technologies more attractive. Drillers, by contrast, are the first or second most in-demand

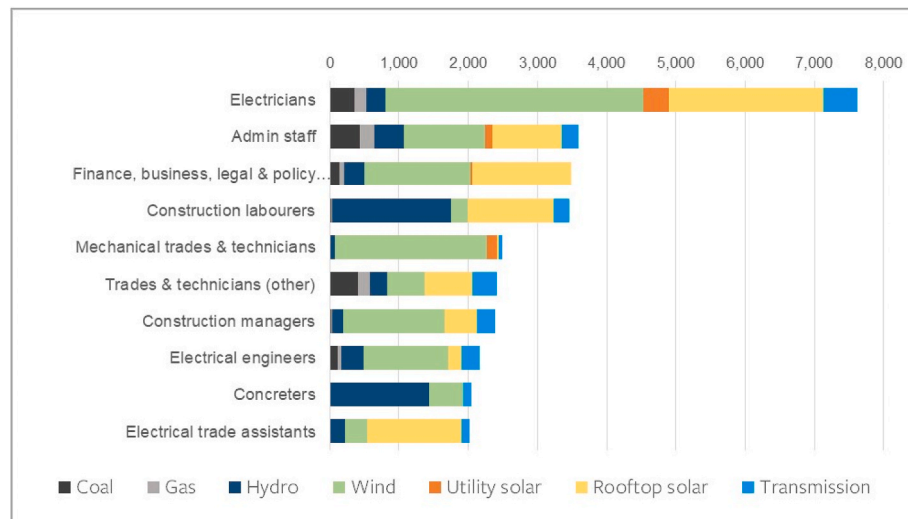


Fig. 2. In-demand occupations for the Australian Integrated System Plan, in a peak year (2029).

Table 8

Three most in-demand occupations for each technology in the peak year (2029).

Coal	Gas	Hydro	Wind	Utility solar	Rooftop solar	Transmission
drillers	metal trades	construction labourers	electricians	electricians	electricians	line workers
metal trades	drillers	concreters	mechanical trades & technicians	operations & production managers	finance, business, legal & policy	riggers & dogmen
admin staff	admin staff	truck drivers	finance, business, legal & policy	executives & senior managers	electrical trade assistants	electricians

occupation for both coal and gas (see Table 9).

An occupational breakdown for the average employment between 2025 and 2040 by technology across all project phases is presented in Table 8. Note that the breakdown will vary according to the period, as it varies by project phase. In times of intense construction, construction occupational shares will dominate the occupational composition, while the operations and maintenance indicators will increase in relative importance as the fleet of renewable generation technology increases.

4. Limitations

Employment factors were produced based on 2020 industry surveys for solar, wind, pumped hydro, conventional hydro, and batteries and 2021 surveys for transmission conventional. There are many sources of error in these indicators. The surveys may be out of date, and the use of cost declines as proxies for changes in employment intensity may either under or overestimate the change. Any single project may have characteristics or challenges which mean that these averaged employment indicators vary from actual employment.

The renewable generation technologies other than conventional hydro are at relatively early stages of industrial development compared to unabated fossil fuel generation technologies. While indicators have been updated by use of cost declines, this will not capture step changes in industry practice, and the surveys should be repeated from time to time.

The use of batteries in stationary energy applications is an immature industry undergoing rapid growth, and the surveys were undertaken at a very early stage of industry development. While the indicators for large batteries have been updated from literature in the intervening period, a re-survey of this industry is overdue.

Application of the indicators in different jurisdictions needs to be undertaken with caution, and results seen as indicative unless

verification of the indicators is undertaken. All technologies are more labour intensive when first implemented in a particular jurisdiction. Employment in distributed technologies will be influenced by housing type and local regulation, as well as by labour productivity. Employment in coal mining is highly dependent on local resources, work practices, and development stage, as extraction can vary from deep mining to highly automated open cast. Employment in large-scale technologies may be more similar, particularly in economies at a similar development stage.

Detailed occupational shares are not available for batteries or offshore wind employment, so occupational analysis is incomplete.

5. Discussion

This research has presented a suite of updated gross employment factors and the associated occupational shares for Australia for a range of energy technologies. These may be used alongside energy scenarios to forecast overall energy workforce requirements.

While overall employment projections are interesting, at this stage of the energy transformation, detailed occupational breakdowns are needed. Training bodies, students, workers, and others who are making the choices that will enable workforce change and growth to make the energy transition a reality need to know not just how many jobs, but the specific occupations needed. Projections can be compared to supply estimates to determine if the business-as-usual labour supply is likely to be adequate. In regional areas, comparing forecasts to the current labour supply can illustrate some of the challenges in gaining social license, as requirements may exceed regional labour supply by a considerable margin, leading to a lack of local benefit [45]. Understanding these challenges allows policy solutions, such as targeted training programmes, which may both alleviate the shortages and increase the local benefits.

Table 9

Average occupational shares by technology, Australian National Electricity Market 2025–2040.

		Coal	Gas	Hydro	Pumped hydro	Wind	Utility solar	Rooftop solar	Transmission	TOTAL
MANAGERS	Executives & senior	1.8 %	1.5 %	1.3 %	2.2 %	29.5 %	17.7 %	44.8 %	1.3 %	1470
	Business development	2.4 %	2.0 %	4.5 %	8.9 %	7.5 %	0.1 %	74.4 %	–	588
	Business	4.7 %	3.8 %	9.3 %	4.7 %	25.2 %	20.3 %	31.0 %	0.9 %	1092
	Construction	1.0 %	0.9 %	1.2 %	3.4 %	41.5 %	8.1 %	35.2 %	8.7 %	1283
	Engineering	22.8 %	19.6 %	22.5 %	8.0 %	–	–	–	27.2 %	114
	Operations & production	3.4 %	2.0 %	1.0 %	0.6 %	72.5 %	20.5 %	–	–	1337
	Logistics & procurement	2.3 %	1.8 %	0.1 %	1.4 %	42.1 %	3.8 %	46.9 %	1.6 %	606
	Managers (other)	51.3 %	48.7 %	–	–	–	–	–	–	210
PROFESSIONALS	Finance, business, legal & policy	3.0 %	2.1 %	3.5 %	2.8 %	38.8 %	2.5 %	47.4 %	–	2923
	Professionals (other)	5.6 %	5.2 %	2.0 %	3.5 %	19.0 %	0.7 %	60.9 %	3.1 %	1173
	Civil engineers	3.3 %	2.9 %	8.5 %	33.1 %	36.0 %	7.4 %	–	8.7 %	408
	Electrical engineers	6.5 %	6.0 %	7.2 %	7.6 %	41.2 %	4.1 %	17.2 %	10.3 %	1086
	Mechanical engineers	19.4 %	17.2 %	11.1 %	24.2 %	19.6 %	8.5 %	–	–	275
	Other engineers	44.1 %	27.9 %	14.4 %	5.1 %	–	–	–	8.4 %	96
	Environment, health & safety	3.7 %	3.1 %	7.7 %	–	63.0 %	7.6 %	12.0 %	2.9 %	1145
	IT professionals	12.6 %	11.0 %	–	–	–	–	75.0 %	1.4 %	268
	Telecommunications engineer	1.9 %	1.7 %	34.6 %	16.5 %	38.5 %	1.9 %	–	4.9 %	215
	Community engagement	–	–	0.5 %	6.6 %	85.0 %	0.2 %	–	7.8 %	301
	Geotechnical engineers	–	–	–	–	–	–	–	100 %	9
	Structural engineers	–	–	–	–	–	–	–	100 %	45
	Quantity surveyors	–	–	–	–	–	–	–	100 %	7
TRADES & TECHNICIANS	Trades & technicians (other)	15.1 %	11.6 %	0.3 %	3.9 %	12.2 %	6.9 %	40.5 %	9.5 %	1630
	Metal trades	46.5 %	33.5 %	6.7 %	9.8 %	3.4 %	–	–	–	885
	Construction trades	14.9 %	11.0 %	–	–	39.9 %	32.5 %	–	1.6 %	38
	Plumbers	24.7 %	22.9 %	–	–	34.0 %	18.4 %	–	–	33
	Electricians	2.9 %	2.4 %	0.9 %	1.0 %	50.4 %	10.9 %	28.8 %	2.8 %	7542
	Lineworkers	6.0 %	6.0 %	–	–	–	–	–	88.0 %	471
	Telecommunications technician	16.9 %	15.8 %	13.9 %	12.3 %	14.3 %	25.5 %	–	1.1 %	124
	Control room/power plant operators	28.4 %	28.4 %	8.3 %	2.9 %	21.9 %	10.1 %	–	–	836
	Civil engineering technicians	1.0 %	0.9 %	35.0 %	63.1 %	–	–	–	–	87
	Mechanical trades & technicians	0.1 %	0.1 %	2.0 %	0.7 %	82.7 %	12.6 %	1.0 %	0.9 %	2649
	Cable jointers	–	–	–	–	–	–	–	100.0 %	4
ADMIN	Admin staff	8.9 %	7.7 %	5.1 %	3.8 %	28.0 %	10.2 %	32.7 %	3.5 %	2977
MACHINE OPERATORS & DRIVERS	Crane operators	2.2 %	2.2 %	0.7 %	9.2 %	71.4 %	–	–	14.3 %	226
	Machine operators & drivers (other)	32.1 %	20.6 %	3.0 %	1.1 %	–	–	21.6 %	21.6 %	262
	Drillers	61.6 %	27.1 %	–	–	8.0 %	–	–	3.3 %	710
	Earthmoving plant operators	7.3 %	4.3 %	0.5 %	7.2 %	50.2 %	3.6 %	–	26.9 %	520
	Truck drivers	7.0 %	3.0 %	1.4 %	18.4 %	8.4 %	8.5 %	41.6 %	11.7 %	1073
	Storepersons & packers	5.8 %	4.4 %	–	–	–	2.6 %	87.2 %	–	476
LABOURERS	General labourers	30.4 %	27.2 %	1.9 %	0.7 %	10.5 %	4.6 %	24.7 %	0.0 %	230
	Construction labourers	0.7 %	0.5 %	1.7 %	20.1 %	3.9 %	14.8 %	54.1 %	4.2 %	2233
	Steel fixers	12.9 %	12.3 %	1.6 %	21.3 %	28.9 %	–	–	23.0 %	353
	Concreters	0.0 %	0.0 %	4.4 %	59.1 %	27.9 %	–	–	8.5 %	637
	Riggers & dogmen	–	–	0.9 %	12.3 %	30.2 %	4.2 %	–	52.4 %	476
	Electrical trade assistants	–	–	0.3 %	3.6 %	7.6 %	–	85.2 %	3.3 %	1548

Projections are not an exact science, and any projection of the future is almost certainly going to be wrong: the energy scenarios (the primary input) themselves may differ significantly from what transpires. There may be step changes in labour practices arising from automation, which could even be driven by labour shortages. However, the advantage of employment factors is that they can be modified and applied to scenarios as circumstances and information changes. Updated workforce projections can inform plans to expand training where needed and can feed into transition planning for affected communities. This is not just for fossil fuel communities that are losing jobs, but for regions that are gaining significant renewable infrastructure. To develop training programs to increase the skills and employment of local people, ensuring benefits accrue locally, data is needed on which occupations are required and the location.

The indicators have been derived for the Australian industry. These

should be verified for use in other jurisdictions, particularly those which are at a different stage of economic development where construction may be more labour intensive. However, they may be used to gain an initial insight on the scale of employment, particularly for large-scale renewable technologies. The employment intensity of coal production is highly location specific, reflecting the great variation in mining productivity, so fuel indicators should be derived for each country.

Workforce analysis was undertaken for the 2022 Integrated System Plan for Australia [12]. Following stakeholder consultation, the Australian Energy Market Operator determined to publish workforce projections alongside electricity scenarios. This was implemented for the 2024 round of electricity planning, and included a recommendation that the 2026 electricity modelling includes a sensitivity aimed at reducing workforce volatility to give policy makers and other stakeholders information to determine policy priorities [58].

6. Conclusion

This study has presented a suite of gross employment factors and detailed occupational shares which can contribute to detailed workforce projections in the electricity sector. There is an argument for including such projections in all electricity sector planning, to inform policy makers and other stakeholders on both the priorities for workforce development and the risks from potential boom and bust cycles. Projections can inform energy policy, as mechanisms such as auctions can be modified to both mandate minimum standards for training or local inclusion, and to stage development to avoid extreme workforce fluctuations.

While the suite of employment factors presented here are a start, there are growth technologies, in particular batteries, where there is a lack of occupational indicators. Just as importantly, the employment factors presented are for Australia, a high income, highly industrialised country, and indicators may be very different in rapidly developing countries where the bulk of new renewables will be built in the next thirty years.

There is also a need to urgently consider the workforce requirements for the other side of the energy transformation, that is, energy efficiency, demand management, and electrification. The suite of energy employment factors needs to be expanded to include the demand side, those regions currently poorly served, and occupational breakdowns for all the important technologies. This will enable governments, education providers, and industry to undertake the workforce planning to make the energy transition a reality.

CRedit authorship contribution statement

J. Rutovitz: Conceptualization, Methodology, Formal analysis, writing - original & revisions, Supervision. **R. Langdon:** Methodology, Formal analysis, writing - original, data acquisition. **C. Briggs:** Conceptualization, Methodology, data acquisition, Formal analysis, Writing - review & editing. **F. Mey:** data acquisition, Formal analysis. **E. Dominish:** data acquisition, Formal analysis. **K. Nagrath:** Methodology, data acquisition, Formal analysis.

Declaration of competing interest

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Data availability

The data that has been used is confidential.

References

- [1] International Energy Agency. World Energy Outlook 2023. 2023.
- [2] Rutovitz J, Visser D, Sharpe S, Taylor H, Jennings K, Atherton A, et al. E3 Opportunity Assessment: Developing the future energy workforce. 2021.
- [3] Rutovitz J, Dominish E, Downes J. Calculating global energy sector jobs 2015 methodology update. 2015.
- [4] Ram M, Aghahosseini A, Breyer C. Job creation during the global energy transition towards 100% renewable power system by 2050. Technol Forecast Soc Change 2020;151:119682. <https://doi.org/10.1016/j.techfore.2019.06.008>.
- [5] Koasidis K, Nikas A, Van de Ven DJ, Xexakis G, Forouli A, Mittal S, et al. Towards a green recovery in the EU: aligning further emissions reductions with short- and long-term energy-sector employment gains. Energy Pol 2022;171. <https://doi.org/10.1016/j.enpol.2022.113301>.
- [6] Malik A, Bertram C, Kriegler E, Luderer G. Climate policy accelerates structural changes in energy employment. Energy Pol 2021;159:112642. <https://doi.org/10.1016/j.enpol.2021.112642>.
- [7] Buchmayr A, Taelman SE, Thomassen G, Verhofstadt E, Van Ootegem L, Dewulf J. A distance-to-sustainability-target approach for indicator aggregation and its application for the comparison of wind energy alternatives. Renew Sustain Energy Rev 2023;185. <https://doi.org/10.1016/j.rser.2023.113608>.
- [8] Briggs C, Rutovitz J, Jazbec M, Landon R, Nagrath K. Employment and Material Requirements for the Integrated System Plan: Electricity Generation and Transmission. Revision 1. 2022.
- [9] Renew Economy. Big battery storage map of Australia n.d. <https://reneweconomy.com.au/big-battery-storage-map-of-australia/>. [Accessed 13 February 2024].
- [10] Clean Energy Council. Clean Energy Australia Report 2020. 2020.
- [11] Briggs C, Rutovitz J, Dominish E, Nagrath K. Renewable Energy Jobs in Australia: Stage One Prepared for the Clean Energy Council by UTS Institute for Sustainable Futures. 2020.
- [12] Rutovitz J, Langdon R, Mey F, Briggs C. The Australian electricity workforce for the 2022 integrated system plan: projections to 2050. Revision 1. 2023.
- [13] Briggs C, Rutovitz J, Jazbec M, Langdon Rusty, Nagrath K. Market Capacity for electricity generation and transmission projects A report from Infrastructure Australia's Market Capacity Program. 2021.
- [14] Aurecon. Aurecon 2022 Cost and Technical Parameter Review 2022.
- [15] Global Wind Energy Council. Global Wind Workforce Outlook 2023.
- [16] International Renewable Energy Agency. Renewable energy and jobs. Annu Rev 2023;2023.
- [17] Graham C, Hayward P, Foster J, Havas J. GenCost 2021-22 Final report 2022.
- [18] Australian Energy Market Operator. 2021-22 Inputs and Assumptions Workbook 2022.
- [19] Briggs C, Hemer M, Howard P, Langdon R, Marsh P, Teske S, et al. Offshore wind energy in Australia. Launceston; 2021.
- [20] National Renewable Energy Laboratory. Jobs and economic development impact model - JEDI. Offshore Wind Tool 2022.
- [21] Rutovitz J, Briggs C, Dominish E, Nagrath K. Renewable energy employment in Australia: methodology. Prepared for the clean energy Council by the Institute for sustainable futures. University of Technology Sydney; 2020. p. 1-42.
- [22] Aslesen S, Vennerød Ø, Erraia J. Outlook for jobs creation in European battery industry 2022.
- [23] Steen M, Lebedeva N Di, Persio F, Boon-Brett L. EU competitiveness in advanced Li-ion batteries for E-mobility and stationary storage applications-opportunities and actions. <https://doi.org/10.2760/75757>; 2017.
- [24] Sandoval B, Anderson P. Gigafactory #1 employment ramp empowering success. Las Vegas; 2018.
- [25] Faraday Institution. Annual Gigafactory Study. UK electric vehicle and battery production potential to 2040. 2020.
- [26] Thielmann A, Neef C, Hettesheimer T, Ahlbrecht K, Ebert S. Future Expert Needs in the Battery Sector 2021. Berlin.
- [27] Eckl-Dorna VW. Hier sollen die Akkus für deutsche E-Autos entstehen. Manag Mag 2021.
- [28] Berenson L, Blaurock J, Kusch PW, Reichel F. BVES Sector Analysis 2022 2022. Berlin.
- [29] Usai L, Lamb JJ, Hertwich E, Burheim OS, Strømman AH. Analysis of the Li-ion battery industry in light of the global transition to electric passenger light duty vehicles until 2050. Environ Res: Infrastructure and Sustainability 2022;2:011002. <https://doi.org/10.1088/2634-4505/ac49a0>.
- [30] Harloff T, Hebermehl G, Wittich H, Schönfeld M, Leicht L. Northvolt treibt Bau deutscher Gigafactory voran. Auto motor sport. 2023.
- [31] Ernst & Young. Solar PV Jobs & Value Added in Europe 2017.
- [32] Solar Energy Industries Association. Solar industry research data n.d. <https://www.seia.org/solar-industry-research-data>. [Accessed 12 February 2024].
- [33] International Renewable Energy Agency. Renewable Energy Benefits Leveraging Local Capacity for Solar PV 2017.
- [34] Paidipati J, Chung G, Esposito A, Marty R, Keyzer D, Tegen S. Workforce development for US hydropower. Prepared by navigant consulting, inc. and the national renewable energy laboratory for the. U.S. Department of Energy; 2017.
- [35] National Renewable Energy Laboratory. Jobs and Economic Development Impact (JEDI) Model - Conventional Hydro. 2016.
- [36] Daw J, Stefek J, DeGeorge E, Philip RT, Cardinal A, Parkhill L. U.S. Hydropower Workforce: Challenges and Opportunities. 2019.
- [37] Biggar Economics. The Economic Impact of Pumped Storage Hydro. A report to Scottish Renewables. 2023.
- [38] International Renewable Energy Agency. Renewable Energy Benefits: Leveraging Local Capacity for Onshore Wind. 2017.

- [39] International Renewable Energy Agency. Renewable energy benefits. Leveraging Local Capacity for Offshore Wind; 2018.
- [40] Global Wind Energy Council. Global Wind Workforce Outlook 2023.
- [41] Sylvest T. Socio-economic impact study of offshore wind 2020.
- [42] Knol E, Coolen E. Post 2030 offshore wind employment in the Netherlands First indications of an outlook on direct employment regarding construction and operations & maintenance phases. 2020.
- [43] Australian Energy Market Operator. 2024 Integrated System Plan ISP for the National Electricity Market. 2024.
- [44] Rutovitz J, Lara H, Gerrard E, Briggs C, Langdon R, Tahir F, et al. The Australian Electricity Workforce for the 2024 Integrated System Plan: Projections to 2050. 2024.
- [45] Briggs C, Atherton A, Gill J, Langdon R, Rutovitz J, Nagrath K. Building a 'Fair and Fast' energy transition? Renewable energy employment, skill shortages and social licence in regional areas. Renewable and Sustainable Energy Transition 2022;2. <https://doi.org/10.1016/j.rset.2022.100039>.